

Reading, Not Reasoning: A Causal Audit of Internalizing Agentic Visual Reasoning into Single-Forward VLMs

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Abstract

A popular recipe for efficient multimodal reasoning is to *internalize* an agent’s multi-step trajectory—reflection, orchestration, tool use—into a single tool-free forward pass of a small vision–language model (VLM). We ask two questions this literature rarely separates: **(1)** where does agentic reasoning over *fixed* visual evidence have any headroom to internalize, and **(2)** when distillation does lift accuracy, is the internalized chain-of-thought (CoT) causally load-bearing? We build a variance-gated map over 3 datasets (NExT-GQA, CLEVRER, ChartQA), 4 model scales (4B–32B), and *three vision-encoder paradigms* (Qwen3-VL tiling-ViT, InternVL3 fixed-resolution ViT, Penguin-VL’s LLM-based encoder). The map is two-regime and stark: in the frames-visible *video* regime, no agentic or reflective method reliably beats a single forward pass in *any* cell—a multi-seed gate retracts our own initially promising +8% as run variance—and the only statistically reliable agentic effects are *negative* (orchestration corrupts answers by 8–16%). A *reasoning-bound* regime appears only once a strong base solves perception (32B on static charts). In that regime we obtain a genuinely significant internalization gain (8B ChartQA: .617→.733, McNemar $p=.023$) and a clean 32B QLoRA gain (.700→.767, 4 gained / 0 lost). But a Lanham-style causal probe over a 2×2 of $\{8B, 32B\} \times \{\text{image present, image masked}\}$ shows the internalized CoT is *not load-bearing under real inference*: with the chart visible, corrupting a CoT intermediate flips the answer no more than shuffling the CoT (32B: 2.2% $\approx$ 2.2%); a latent reasoning chain surfaces only when the image is masked. The stronger base bypasses its CoT *more*, independently reproducing “larger $\Rightarrow$ less faithful CoT” in a VLM. We conclude that trajectory distillation transfers *perception/reading*, not load-bearing reasoning—and that accuracy-only evaluations of “internalized reasoning” systematically overclaim.

1 Introduction

Agentic VLM systems—iterative frame retrieval, self-reflection, critic-orchestrated re-reading, visual tool use—report gains over single-pass inference, motivating a distillation step: convert the trajectory into a CoT and fine-tune a small VLM to reproduce it tool-free at inference (Adhikari and Lapata, 2026; Hu et al., 2024; STAR authors, 2026). Two assumptions are implicit and largely untested:

A1 (headroom). Holding visual evidence fixed, the agent’s *reasoning* (not its frame selection) beats the base model’s single forward pass, so there is something type-1 (single-forward-doable) to internalize.

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A2 (faithfulness). When distillation lifts accuracy, the student’s emitted CoT is the causal mechanism of the gain.

We test both. For A1 we run a multi-seed, paired-bootstrap **variance gate** over every (dataset $\times$ model $\times$ agentic-method) cell, with a label audit, a frames-visible partition, and an explicit power analysis. For A2 we run a causal CoT-intermediate probe (CORRUPT one numeric intermediate vs. SHUFFLE the whole CoT, force-continue) on the SFT’d students, with an **image-masked control** that disambiguates “no reasoning learned” from “reasoning learned but bypassed.”

Contributions.

1. A statistically gated **two-regime map**: video QA with the evidence on-screen is perception-/selection-bound at every scale (4B/8B/32B) and across three encoder paradigms; agentic nets are within run variance everywhere, and the only reliable effects are negative. The reasoning-bound regime exists, but only where a strong base has already solved perception (32B + static charts).
2. A cautionary **retraction case study**: our own single-seed “+8% orchestrated reflection” headline washes out to +0.9% with sign flips across 10 seeds—single-seed agentic gains on $n \approx 50$ benchmarks are not evidence.
3. Two **positive internalization proofs-of-concept** in the reasoning-bound cell (8B: +11.7%, CI excludes 0, McNemar $p=.023$; 32B QLoRA: +6.7%, 4/0 gain/loss, underpowered $p=.125$), with per-epoch held-out selection shown to be load-bearing methodology.
4. The first (to our knowledge) **causal CoT-intermediate audit of an internalized, no-tool, single-forward VLM**, with an image-masked control. Verdict: the distilled gain is perception/reading transfer; the CoT is bypassed under real inference, more so at larger scale.

2 Related Work

Trajectory/agent distillation. Visual Program Distillation (Hu et al., 2024) distills visual-program traces into single-pass VLMs and describes the result as “faithful” on accuracy evidence alone; PEARL (Adhikari and Lapata, 2026) internalizes expert tool-use trajectories in latent space, avoiding tool invocation at inference; STAR (STAR authors, 2026) distills multi-agent trajectories (planning, tool use, self-reflection) into one student—all evaluated by accuracy, none causally probing the student’s CoT. DualDistill’s Agentic-R1 (Du et al., 2025), Agent0-VL (Liu et al., 2025), Pixel-Reasoner (Su et al., 2025), and DeepEyes (Zheng et al., 2025) keep tool invocation at inference and are therefore outside the single-forward setting.

Causal probing of CoT. Lanham et al. (2023) introduce corrupt/truncate interventions on text-LLM CoT and find larger models produce less faithful CoT. CodeV (Hou et al., 2025) is the closest work: it applies causal perturbations (Mask/Noise/Random/Empty)—but to *tool-output images* of a tool-using agent, i.e., perception inputs, not CoT intermediates of an internalized model. The intersection we occupy—Lanham-style intermediate corruption on a distilled no-tool VLM, plus a perception-vs-reasoning regime boundary and an image-masked control—appears unoccupied as of June 2026.

Failure taxonomies. Diagnostic studies report that most VLM “reasoning” failures are perceptual: VRIQ (Khezresmaeilzadeh et al., 2026) attributes 56% of failures to perception alone, 43% to perception-plus-reasoning, and only 1% to reasoning alone—consistent with our regime-1 mechanism.

3 Experimental Setup

Tasks. *NExT-GQA* (Xiao et al., 2024) (short-video MCQ with grounding ground truth; 16 uniformly sampled frames; the **frames-visible regime**—we verify all 50 pilot cases are EVIDENCE-IN, i.e., the uniform sample hits the GT window), *CLEVRER* (Yi et al., 2020) (collision reasoning, 70 cases), and *ChartQA* (Masry et al., 2022) (static-chart numeric QA, 60 held-out test items). A text-only LLM label audit on NExT-GQA yields 47 clean / 3 ambiguous / 0 wrong-gold; all video gates run on the $\text{CLEAN} \cap \text{EVIDENCE-IN}$ set ($n=47$).

Models. Qwen3-VL 4B / 8B / 30B-A3B (MoE) / 32B (dense) (Bai et al., 2025); InternVL3-8B (Zhu et al., 2025) (fixed-resolution ViT; cross-family control); Penguin-VL 2B/8B (Zhang et al., 2026) (LLM-based vision encoder with temporal token compression; cross-*paradigm* control). All models are served locally; the orchestrator/critic is a text-only LLM, never a stronger VLM teacher for the video gates.

Agentic methods (frames held fixed = the type-1 isolation). SELF-REFLECT (2-turn tool-free self-re-examination over the same frames); ORCH-BLIND (a text critic poses sub-questions, the VLM is the eyes, the critic integrates); ORCH-SIGHTED (a 32B-VLM critic that can re-read the image). Frame retrieval is deliberately excluded: it changes the evidence (type-2) and cannot be internalized into a fixed-input forward pass.

Statistics. K seeds per cell ($K=10$ for 4B/8B NExT, 6 for 32B, 3–5 elsewhere); paired bootstrap 95% CIs on the net (gain – loss) over cases; McNemar exact tests for SFT deltas; and a power table: at $n=47$ the minimum detectable net at 80% power is $\approx 25\%$ —so all $|\text{net}| < 9\%$ nulls are reported as “no reliable effect, underpowered,” with cross-seed sign flips as the power-independent evidence.

4 Result I: The Two-Regime Map

Table 1 summarizes 13 cells.

(a) **No cell shows reliable agentic headroom; the reliable effects are negative.** Across all cells, no agentic/reflective method beats single-forward outside run variance. Three cells show statistically reliable *harm* from orchestration: 4B-ChartQA -12.2% , InternVL3-NExT -16.3% (all seeds negative), Penguin-2B-ChartQA -7.8% . Per-case output dumps show two mechanisms: leading sub-questions make the suggestible VLM emit contradictory new reads (video), and decompose-and-reintegrate corrupts an answer whose sub-reads were correct (charts). Multi-step orchestration adds more failure points than it fixes.

Table 1: The variance-gated map. “Best agentic net” is the best of SELF-REFLECT/ORCH-BLIND/ORCH-SIGHTED vs. free-form, pooled over K seeds (paired bootstrap). No cell shows reliable positive headroom; reliably negative cells are **bold**. R1 = perception/selection-bound; R2 = reasoning-bound.

Dataset $\times$ model	Free acc	Best agentic net	Verdict	Regime
NExT 4B / 8B / 32B	.74 / .66 / .70	−.045 / +.060 / +.039	within-var. ($K=10/10/6$)	R1
NExT InternVL3-8B	.57	orch −.163	rel. neg. (CI $[-.32, -.02]$)	R1 (family)
NExT Penguin 2B / 8B	.53 / .43 [†]	+ .050 / +.128	within-var. ($K=3$)	R1 (paradigm)
CLEVRER 4B–32B	.43–.50	≈ 0	$\approx$ chance; tracking wall	R1
ChartQA 4B / 8B	.67 / .62	+ .023 / +.012 (4B orch −.122)	within-var. / rel. neg.	R1 (reading)
ChartQA Penguin 2B / 8B	.82 / .80	−.078 / −.006	within-var. ($K=3$)	R1 (enc. solves)
ChartQA 32B	.80	+ .017	within-var.	R2

[†]Penguin-8B on NExT ran frame-starved at 2 frames (memory limit); not comparable to 16-frame cells.

(b) The retraction. Our single-seed sweep had one positive headline: 8B-NExT orchestrated reflection +8%. Ten seeds give per-seed nets of +.09, −.04, +.11, +.04, +.02, .00, +.06, −.06, −.06—pooled +**0.9%**, sign-flipping. The variance source is the critic’s stochastic sub-questions, not the VLM’s decoding (SELF-REFLECT decode variance ≈ 0 on MCQ). Single-seed agentic gains at this n are noise.

(c) Why regime 1 is perceptual: converging probes. On the 18 8B-NExT free-form errors, giving the model the *ground-truth* frames at maximum resolution recovers only 4/18 (22%); a near-oracle (32B captions of GT frames $\rightarrow$ text-LLM reasons) recovers 6/18 (33%). Two-thirds of the failure surface is unreachable by any reasoning-over-fixed-frames intervention. The sighted-critic ablation sharpens the mechanism: a 32B sighted critic trends positive on every seed on *charts* (+.070, where the wall is re-readable values) but not on *video* (+.060, CI crossing 0, where the wall is temporal localization that re-reading the same 16 frames cannot fix).

(d) The encoder-paradigm result. Penguin-VL’s LLM-based encoder lifts small-model chart reading to **.82 (2B)**—matching Qwen-32B (.80) and far above Qwen-8B (.62)—proving the small-model chart-perception wall is an *encoder artifact, not a capability limit*. Yet the temporal-video wall is unmoved (Penguin-2B NExT .53) and agentic nets stay ≤ 0 . The regime structure is a property of the *task*, robust across three encoder paradigms.

5 Result II: Internalization Works in Regime 2—by Accuracy

In the one reasoning-bound cell, we run the actual distillation. A 32B teacher generates step-by-step chart CoTs on ChartQA-train; 150 rationales pass an answer-consistency filter; we LoRA-SFT the student (vision tower frozen) and evaluate single-forward, tool-free on the 60 disjoint test items.

- **8B student:** .617 $\rightarrow$ **.733**, net +**0.117**, bootstrap CI $[+.050, +.200]$ excludes 0, gain 7 / lost 0, McNemar $p = .023$. A real, significant single-forward internalization gain.
- **32B student (QLoRA, NF4 + $r=16$):** base-with-CoT-prompt .700 $\rightarrow$ **.767** at the epoch-1/2 peak, gain 4 / lost 0, but McNemar $p = .125$ (underpowered at $n=60$).

Two methodological notes generalize. (i) Train loss is monotone-meaningless for a 150-example LoRA—test accuracy peaks at epoch 1–2 and *declines* by epoch 4 (.717, losses appear), so **per-epoch held-out selection is load-bearing**; a blind 3-epoch recipe picks a strictly worse checkpoint. (ii) The CoT prompt itself *costs* the untrained 32B $\sim 10\%$ (free-form .80 vs. CoT-prompted

Table 2: Causal CoT-intermediate probe on the SFT’d students. “Flip” = the final answer changes after the intervention. A load-bearing chain requires CORRUPT $\gg$ SHUFFLE.

Base	Condition	n	CORRUPT-flip	SHUFFLE-flip	Reading
8B	image present	48	16.7%	27.1%	$\text{SHUFFLE} \geq \text{CORRUPT} \rightarrow$ chain not load-bearing
8B	image masked	48	37.5%	27.1%	$\text{CORRUPT} > \text{SHUFFLE} \rightarrow$ latent chain exists
32B	image present	46	2.2%	2.2%	CoT ignored; model re-reads chart
32B	image masked	46	10.9%	6.5%	$\text{CORRUPT} > \text{SHUFFLE} \rightarrow$ latent chain exists

.70); the SFT mostly recovers that penalty and does not exceed the base’s best free-form .80. The clean $\pm$ adapter contrast is the correct causal isolation of the LoRA, but headline “+X% from CoT distillation” numbers that compare against a CoT-prompted base deserve scrutiny everywhere.

6 Result III: The Causal Probe—the Gain Is Reading, Not Reasoning

For every test case the student answers correctly with an emitted CoT, we intervene Lanham-style: CORRUPT one numeric intermediate (a targeted hit on the arithmetic chain) vs. SHUFFLE the CoT sentences (a control that destroys prose coherence but no specific value), then force-continue to a final answer. If the chain is load-bearing, CORRUPT-flips $\gg$ SHUFFLE-flips. We run the full 2×2 of $\{8B, 32B\} \times \{\text{image present} = \text{real inference}, \text{image masked} = \text{CoT is the only information}\}$ in one in-process harness (Table 2).

Three findings:

1. **Under real inference (image present), the internalized CoT is not load-bearing at either scale.** The 8B is more disturbed by scrambled prose than by a falsified number; the 32B is disturbed by neither ($\approx 2\%$)—it silently re-derives the answer from the chart. The accuracy gains of §5 are *perception/reading transfer*: the teacher’s trajectory taught the student *where and how to read*, not an arithmetic chain the answer rides on.
2. **The stronger the base, the more it bypasses its own CoT** (CORRUPT-flip 2.2% vs. 16.7%). This independently reproduces Lanham et al. (2023)’s “larger models produce less faithful CoT” in a VLM, and identifies the mechanism: a perceptual re-read shortcut that a stronger vision tower can lean on.
3. **The image-masked control is essential.** Masking flips both scales to CORRUPT $>$ SHUFFLE: a usable latent chain *was* internalized, but is only consulted when the perceptual shortcut is removed. Without this control, the present-condition $\approx 0\%$ would be misread as “no reasoning learned”; the truth is “reasoning learned but bypassed.” This also disposes of the objection that our corruption is too weak to detect any chain.

7 Discussion and Limitations

Implications. (i) Claims of “internalizing agentic reasoning” need a causal audit—accuracy deltas, even significant ones, are compatible with pure perception transfer, and in our engineered best-case regime-2 cell that is exactly what they were. (ii) The agent’s real edge on video is *perception routing* (frame/region selection)—which is type-2 by construction and cannot be internalized into a fixed-input forward pass; text-level reflection over a fixed read cannot repair perceptual error

and reliably amplifies it. (iii) Single-seed evaluation of agentic methods on $n \approx 50$ benchmarks is uninterpretable; report seeds and paired CIs.

Limitations. Sample sizes are small ($n=47$ video / $n=60$ charts; minimum detectable effect $\approx 25\%$ on video at 80% power)—regime-1 nulls are “no reliable effect, underpowered,” with cross-seed sign flips as the power-independent evidence; scaling n is the first next step. The 32B student is QLoRA on an NF4 base (a bf16 full-precision replication is hardware-deferred). The teacher pool is one family (Qwen3-VL); the cross-family/paradigm controls cover students, not teachers. CoT corruption targets numeric intermediates; richer interventions (truncation, paraphrase, step deletion) would tighten the audit. Penguin-8B’s video cell is frame-starved and not comparable.

8 Conclusion

We set out to compress tool-scaffolded visual reasoning into one forward pass and to verify it causally. The defensible result is a boundary: **where the frames are already visible, there is nothing reliable to internalize—the bottleneck is perception, and agentic reflection is at best noise and at worst reliably destructive.** Where a strong base clears perception, distillation lifts accuracy significantly—but the causal probe shows the student learned to *read* like its teacher, not to *reason* like it, and the stronger the student, the more thoroughly it ignores the chain it was taught to emit. Reasoning that survives a causal audit remains, on this evidence, un-internalized.

Reproducibility Statement

All gate numbers regenerate from a fingerprinted, append-only result store (`results.jsonl`) via the released gate/table scripts; SFT, merge, and probe scripts (`poc_*.py`) and per-case output dumps are included. The variance gate, label audit, frames-visible partition, power table, perception-headroom and near-oracle probes, and the 2×2 causal probe each correspond to one script with fixed seeds.

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